NIKHIL KRISHNAPRASAD

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OBJECTIVE

Software Engineer with 3+ years of experience building scalable full-stack applications, AI-powered systems, and cloud-native platforms. Strong background in Java, Python, React, APIs, distributed systems, AI agents, and RAG-based workflows, with experience developing production-ready software solutions and intelligent automation systems. Passionate about building impactful AI products that solve real-world problems through scalable engineering and rapid innovation.

EDUCATION

Master of Science, Computer Science

Western Michigan University, Kalamazoo, MI

GPA: 3.9/4.0 | Expected Dec 2026

Bachelor of Engineering, Computer Science

Visvesvaraya Technological University, Bengaluru, India
2021

EXPERIENCE

Technology and Systems Intern — Room 35, Kalamazoo, MI | Oct 2025 – Apr 2026

- Developed and deployed an AI-powered chatbot and internal automation tools to streamline employee workflows, improve information accessibility, and enhance operational efficiency.
- Built robust Python data pipelines and automation workflows, improving data accuracy by 40% and reducing manual reporting by 30%.
- Owned end-to-end development of dashboards, APIs, and internal software systems from requirements gathering through implementation, deployment, and iteration.
- Designed scalable system architectures, backend services, and APIs supporting AI-powered applications and enterprise workflow integrations.
- Worked closely with stakeholders to prototype solutions, gather feedback, and iteratively improve software systems based on business needs.

Application Engineer — Alstom, Bangalore, India | Sep 2022 – Jun 2024

- Developed and maintained real-time, scalable applications for railway monitoring systems.
- Improved system responsiveness by 20% and data retrieval efficiency by 30% through performance optimization and efficient data handling.
- Identified and resolved system issues through debugging, log analysis, and performance monitoring.
- Collaborated with cross-functional teams to integrate monitoring interfaces, backend services, and real-time data systems.

Product Engineer — Wipro, Bangalore, India | Aug 2021 – Sep 2022

- Developed and maintained scalable full-stack applications, APIs, and user-facing interfaces for secure payment platforms handling 5,000+ daily transactions.
- Designed scalable services and APIs for transaction processing and validation.
- Improved system scalability and performance by 18% through optimized application design and efficient data handling.
- Developed responsive React.js user interfaces and reusable frontend components, improving usability, maintainability, and overall user experience across payment platform workflows.
- Collaborated in an agile environment, participating in code reviews, testing, and deployment cycles.

SKILLS

Programming Languages: Python, Java, JavaScript, C#, C, C++, PHP

AI & Intelligent Systems: LLMs, RAG, Embeddings, Vector Search, AI Agents, Multi-Agent Systems, LangChain, LlamaIndex, AI-Assisted Development, AI-Native Workflows, Prompt Engineering, MCP, RAGAS

Frontend: React.js, Next.js, TypeScript, HTML, CSS

Backend & Systems: Node.js, RESTful APIs, Microservices, Distributed Systems, Real-Time Processing, Event-Driven Architectures, Asynchronous Systems, Webhooks

Databases: SQL, NoSQL, MongoDB, Pinecone, PGVector

Cloud, DevOps & Infrastructure: AWS, Google Cloud (GCP), Docker, Kubernetes, CI/CD

Data & Processing: Apache Spark, PySpark, Data Pipelines

Software Engineering: Data Structures & Algorithms, SDLC, Code Reviews, Debugging, Testing, Monitoring & Logging

Tools: Git, GitHub, Jira

PROJECTS

AI Workspace Platform

- Built a production-style AI platform integrating multi-agent orchestration, conversational memory, MCP-style tool integration, and scalable retrieval pipelines to automate development and knowledge workflows.
- Developed agentic workflows leveraging tool calling, semantic retrieval, and persistent memory to support complex task execution and knowledge management processes.
- Designed evaluation pipelines using prompt engineering, context management, and automated quality checks to improve response reliability and reduce hallucinations.
- Technologies: Python, FastAPI, LangChain, OpenAI APIs, Pinecone, PGVector, Streamlit, Docker

GitHub Repository Intelligence Platform (MCP)

- Built an MCP-enabled repository intelligence platform capable of analyzing GitHub repositories, performing semantic code search, and answering repository-specific questions using LLMs.
- Integrated MCP-based tool calling, structured context sharing, and retrieval pipelines to enable intelligent code understanding, repository exploration, and agent-driven repository interactions.
- Designed scalable APIs and repository ingestion workflows, improving context relevance, retrieval accuracy, and response quality.
- Technologies: Python, FastAPI, MCP, OpenAI APIs, LangChain, GitHub API, Vector Databases

AI Clinical Assistant System (RAG-based)

- Built a RAG-powered clinical assistant integrating embeddings, vector search, and context-aware retrieval to generate grounded healthcare responses.
- Containerized application services using Docker and deployed scalable workloads in cloud-based environments.
- Technologies: Python, LangChain, Vector Databases, Docker, AWS, Kubernetes

Smart Interview Preparation System

- Developed a real-time interview simulation platform with interactive frontend components, modular APIs, and scalable evaluation workflows.
- Designed user-centric evaluation workflows enabling continuous improvements through feedback analysis and performance metrics.
- Built automated assessment features using regex-based processing and finite state automata for structured response evaluation.
- Technologies**:** Python, Gradio, React.js, REST APIs, Regex Processing, Finite State Automaton